PaLI-3 Vision Language Models: Smaller, Faster, Stronger

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Abstract

This paper presents PaLI-3, a smaller, faster, and stronger vision language model (VLM) that compares favorably to similar models that are 10x larger. As part of arriving at this strong performance, we compare Vision Transformer (ViT) models pretrained using classification objectives to contrastively (SigLIP) pretrained ones. We find that, while slightly underperforming on standard image classification benchmarks, SigLIP-based PaLI shows superior performance across various multimodal benchmarks, especially on localization and visually-situated text understanding. We scale the SigLIP image encoder up to 2 billion parameters, and achieves a new state-of-the-art on multilingual cross-modal retrieval. We hope that PaLI-3, at only 5B parameters, rekindles research on fundamental pieces of complex VLMs, and could fuel a new generation of scaled-up models.

1 Introduction

The scaling of vision-language models (VLM) to tens and even hundreds of billions of parameters (Chen et al., 2023b; Alayrac et al., 2022; Chen et al., 2023a; Driess et al., 2023) has shown ever-increasing performance. Meanwhile, models at a smaller scale remain critical, as they are more practical to train and serve, more environmentally-friendly, and support faster research cycles for model design.

In the spirit of focusing on smaller-scale modeling, we present PaLI-3, the third-generation family of PaLI (Chen et al., 2023b) models. Using a pretrained backbone with only 5B total parameters, we refine the training recipe and achieve competitive and new state-of-the-art (SOTA) results on various VLM benchmarks. Our new recipe has three main components: contrastive pretraining of image encoder on web-scale image-text data (Zhai et al., 2023), an improved dataset mixture for PaLI multimodal training, and training at higher resolutions.

PaLI-3 achieves new SOTA results on tasks that require visually-situated text understanding and object localization, including eight visually-situated text understanding tasks and the referring expression segmentation task on RefCOCO (Yu et al., 2016), along with strong performance on a wide range of classical vision tasks. As part of this work, we also introduce a SOTA multilingual contrastive vision model scaled to 2B parameters, obtained using the recently-introduced SigLIP recipe (Zhai et al., 2023). Additionally, we perform focused ablation studies to compare the classification pretrained Vision Transformer (ViT) backbones (Dosovitskiy et al., 2021) with contrastively pretrained ones (SigLIP). This further confirms the viability of pretraining visual encoders on noisy web-scale image-text data, as a preferable alternative to training on classification-style data.

Our contributions are summarized as follows:

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PALI-3 视觉语言模型：更小、更快、更强

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摘要

本文介绍了 PaLI-3，这是一个更小、更快且更强的视觉语言模型（VLM），其性能优于参数规模大 10 倍的同类模型。为了实现这一强劲性能，我们比较了使用分类目标预训练的视觉 Transformer（ViT）模型与使用对比学习（SigLIP）预训练的模型。我们发现，尽管基于 SigLIP 的 PaLI 在标准图像分类基准上表现略逊一筹，但它在各种多模态基准上展现出更优的性能，尤其是在定位和视觉情境中的文本理解方面。我们将 SigLIP 图像编码器扩展到 20 亿参数，并在多语言跨模态检索上取得了新的最先进水平。我们希望仅有 50 亿参数的 PaLI-3 能够重新激发对复杂 VLM 基础组成部分的研究，并推动新一代扩展模型的发展。

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1 引言

视觉-语言模型（VLM）扩展到数百亿甚至数千亿参数规模（Chen et al., 2023b; Alayrac et al., 2022; Chen et al., 2023a; Driess et al., 2023）后，性能不断提升。与此同时，较小规模的模型仍然至关重要，因为它们在训练和部署上更具实用性，更环保，并且支持更快的模型设计研究周期。

本着专注于较小规模建模的理念，我们提出了 PaLI-3，即第三代 PaLI（Chen 等，2023b）模型系列。我们使用一个总参数量仅为 5B 的预训练骨干网络，优化了训练方案，并在各种 VLM 基准测试上取得了具有竞争力且新的最先进（SOTA）结果。我们的新方案包含三个主要组成部分：在网络规模的图像-文本数据上对图像编码器进行对比式预训练（Zhai 等，2023）、用于 PaLI 多模态训练的改进数据集混合，以及在更高分辨率下进行训练。

PaLI-3 在需要视觉情境文本理解和目标定位的任务上取得了新的 SOTA 成果，包括八项视觉情境文本理解任务以及 RefCOCO（Yu et al., 2016）上的指代表达式分割任务，同时在广泛的经典视觉任务上也表现强劲。作为这项工作的一部分，我们还引入了一个扩展到 20 亿参数的 SOTA 多语言对比视觉模型，该模型使用最近提出的 SigLIP 方法（Zhai et al., 2023）获得。此外，我们还进行了有针对性的消融研究，以比较分类预训练的视觉 Transformer（ViT）主干网络（Dosovitskiy et al., 2021）与对比预训练的主干网络（SigLIP）。这进一步证实了在嘈杂的网络规模图像-文本数据上预训练视觉编码器的可行性，并表明相较于在分类式数据上训练，这是一种更可取的替代方案。

我们的贡献总结如下：

[⋆]Core contributors, [†]Project leads

1. We compare classification pretrained ViT models (Dosovitskiy et al., 2021) to contrastively pretrained SigLIP models using the PaLI framework (Chen et al., 2023b). We find that the contrastively pretrained models work significantly better for visually-situated text understanding tasks and localization tasks.
2. Our model achieves SOTA performance on 10+ diverse vision-language benchmarks while being 10x smaller in size compared to the current SOTA model Chen et al. (2023a). For understanding visually-situated text, the improvement is by a particularly large margin.
3. Despite not pretraining on any video data, our model achieves new SOTA on several video QA benchmarks, indicative of powerful generalization abilities.
4. We introduce the 2 billion parameter (ViT-G) multilingual SigLIP model trained on WebLI (Chen et al., 2023b), which sets a new state-of-the-art on the multilingual cross-modal retrieval benchmark Thapliyal et al. (2022) across 36 languages.

2 Related Work

Recent large vision language models (VLMs) use pretrained image encoders as part of the larger model, some pretrain it with supervised classification (PaLI (Chen et al., 2023b), PaLI-X (Chen et al., 2023a), PaLM-E (Driess et al., 2023)), some use pretrained CLIP encoders (BLIPv2 (Li et al., 2023), CrossTVR (Dai et al., 2023), ChatBridge (Zhao et al., 2023)) and some with custom multimodal pretraining (Flamingo (Alayrac et al., 2022), BEiT3 (Wang et al., 2022b), CoCa (Yu et al., 2022), SimVLM (Wang et al., 2021b)). In this paper we compare two dominant ways to pretrain image encoders using the PaLI framework: classification pretraining using large weakly labeled datasets (JFT, as in Kolesnikov et al., 2020; Zhai et al., 2022a; Dehghani et al., 2023) and contrastive pretraining on web-scale noisy data (WebLI, as in Zhai et al., 2023).

A recent finding spanning across PaLI (Chen et al., 2023b) and PaLI-X (Chen et al., 2023a) is that scaling up the classification pretrained image encoder seems more promising than was previously believed (Alayrac et al., 2022). Specifically, while classic image-only benchmarks such as ImageNet seem to indicate saturating performance from scaling pretraining of image-only models (Zhai et al., 2022a), PaLI shows that by scaling up the vision encoder from ViT-G (2B) to ViT-e (4B), the improvements on VL tasks are more noticeable than on ImageNet. PaLI-X further scaled up both the vision and language components, showing that these larger image encoders keep bringing benefit when plugged into large VLMs. This finding suggests that there is more to be found regarding the pretraining of image encoders in the context of VLMs, which may lead to different conclusions when looking at VLM tasks as compared to of "pure" vision tasks. In this paper, we dive into the impact of the image encoder for VLMs, by directly comparing classification pretrained vision models to contrastively pretrained ones, and reveal that the latter are vastly superior on a variety of tasks, especially localization and visually-situated text understanding.

One can split multimodal understanding capabilities into largely two categories: natural scene understanding (captioning, VQA, object detection/localization), and visually-situated text understanding (document and infographics QA). These groups of tasks require different granularity of understanding, and previous VLMs have largely focused on one type of tasks, leading to their training recipes being attuned to that type of tasks. For example PaLI-17B (Chen et al., 2023b) and Pix2struct (Lee et al., 2022) showed strong performance only on one of the two categories, respectively. The recent PaLI-X (Chen et al., 2023a) achieves SOTA performance on both categories, based on an improved OCR-related training recipe, and a significantly larger 55B parameter model. In this work, we combine the advantage of contrastively-pretrained ViT and a further improved and balanced training recipe into PaLI-3, and demonstrate that SOTA level performance on both the above categories of multimodal understanding is achievable even at 5B parameter scale.

3 Model

3.1 Architecture

On a high level, the architecture follows Chen et al. (2023b;a): a ViT encodes the image into tokens which, together with text input (the question, prompt, instruction), are passed to an encoder-decoder transformer (Vaswani et al., 2017) that generates a text output.

1. 我们使用 PaLI 框架（Chen et al., 2023b），将分类预训练的 ViT 模型（Dosovitskiy et al., 2021）与对比式预训练的 SigLIP 模型进行比较。我们发现，对比式预训练模型在视觉情境文本理解任务和定位任务上表现显著更好。2. 与当前 SOTA 模型 Chen et al. (2023a) 相比，我们的模型规模小 10 倍，同时在 10+ 个多样化的视觉-语言基准上实现了 SOTA 性能。对于视觉情境文本理解，其提升幅度尤其显著。3. 尽管没有在任何视频数据上进行预训练，我们的模型仍在多个视频问答基准上取得新的 SOTA，显示出强大的泛化能力。4. 我们推出了在 We-bLI（Chen et al., 2023b）上训练的 20 亿参数（ViT-G）多语言 SigLIP 模型，该模型在涵盖 36 种语言的多语言跨模态检索基准 Thapliyal et al. (2022) 上创下了新的最先进水平。

2 相关工作

近期的大型视觉语言模型（VLM）将预训练图像编码器作为更大模型的一部分：有些通过有监督分类对其进行预训练（PaLI（Chen et al., 2023b）、PaLI-X（Chen et al., 2023a）、PaLM-E（Driess et al., 2023）），有些使用预训练的 CLIP 编码器（BLIPv2（Li et al., 2023）、CrossTVR（Dai et al., 2023）、ChatBridge（Zhao et al., 2023）），还有一些采用定制的多模态预训练（Flamingo（Alayrac et al., 2022）、BEiT3（Wang et al., 2022b）、CoCa（Yu et al., 2022）、SimVLM（Wang et al., 2021b））。在本文中，我们使用 PaLI 框架比较了两种主流的图像编码器预训练方式：基于大规模弱标注数据集的分类预训练（JFT，如 Kolesnikov et al., 2020；Zhai et al., 2022a；Dehghani et al., 2023）以及基于网络规模噪声数据的对比预训练（WebLI，如 Zhai et al., 2023）。

PaLI（Chen 等，2023b）和 PaLI-X（Chen 等，2023a）的一项近期发现是，扩大经过分类预训练的图像编码器规模似乎比此前认为的更有前景（Alayrac 等，2022）。具体而言，虽然 ImageNet 等经典的仅图像基准似乎表明，仅图像模型的预训练规模扩展会出现性能饱和（Zhai 等，2022a），但 PaLI 表明，通过将视觉编码器从 ViT-G（2B）扩大到 ViT-e（4B），在视觉-语言（VL）任务上的提升比在 ImageNet 上更为显著。PaLI-X 进一步扩大了视觉和语言组件的规模，表明这些更大的图像编码器在接入大型视觉-语言模型（VLM）时仍能持续带来收益。这一发现表明，在 VLM 语境下，关于图像编码器的预训练还有更多值得探索的内容；与考察“纯”视觉任务相比，在考察 VLM 任务时可能会得出不同结论。本文深入研究图像编码器对 VLM 的影响，直接比较经过分类预训练的视觉模型与经过对比预训练的视觉模型，并揭示后者在多种任务上具有显著优势，尤其是在定位和视觉情境文本理解方面。

可以将多模态理解能力大致分为两类：自然场景理解（图像描述、VQA、目标检测/定位）和视觉情境中的文本理解（文档和信息图表问答）。这两类任务需要不同粒度的理解，而以往的 VLM 大多专注于其中一种任务类型，导致其训练方案也针对该类任务进行了调优。例如，PaLI-17B（Chen et al., 2023b）和 Pix2struct（Lee et al., 2022）分别只在这两类任务中的一类上表现出强劲性能。近期的 PaLI-X（Chen et al., 2023a）基于改进的 OCR 相关训练方案以及规模显著更大的 55B 参数模型，在这两类任务上都取得了 SOTA 性能。在本工作中，我们将对比预训练 ViT 的优势与进一步改进且更加均衡的训练方案结合到 PaLI-3 中，并证明即使在 5B 参数规模下，也可以在上述两类多模态理解任务上达到 SOTA 水平的性能。

3 模型

3.1 架构

从高层次来看，该架构遵循 Chen 等人（2023b；a）：ViT 将图像编码为标记，这些标记与文本输入（问题、提示、指令）一起传递给编码器-解码器 Transformer（Vaswani 等人，2017），从而生成文本输出。

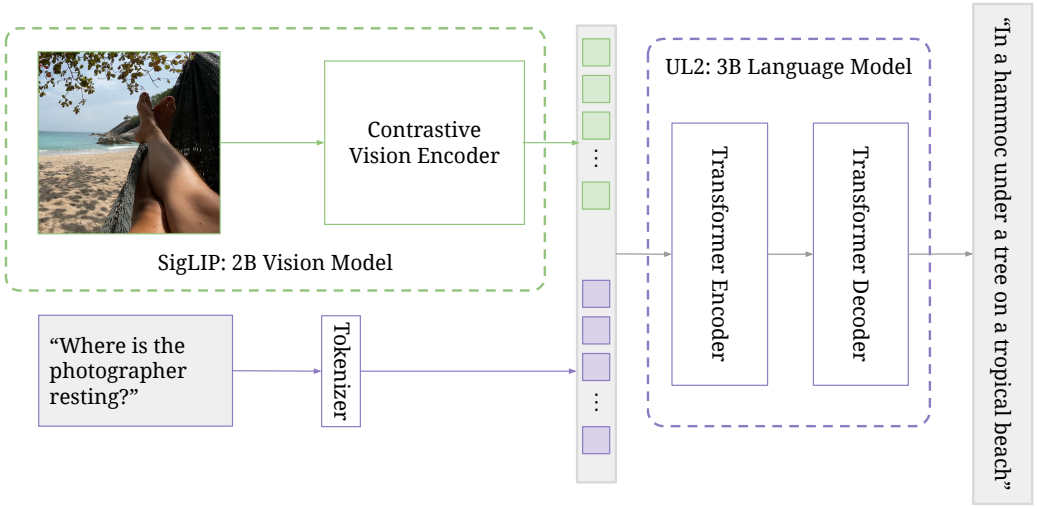


Figure 1: Overview of the PaLI-3 (5B) model: images are encoded into visual tokens individually by the contrastively pretrained 2B SigLIP vision model. Along with a query, these visual tokens are passed to an 3B encoder-decoder UL2 Transformer which produces the desired answer. In such a setup, a contrastively pretrained model provides significantly more useful tokens than one classification pretrained model as in previous PaLI models.

Visual component The vision backbone of PaLI-3 is initialized from a contrastively pretrained ViT-G/14[2] (Zhai et al., 2022a) model (approx. 2B parameters) using the SigLIP (Zhai et al., 2023) training recipe. In short, an image embedding ViT-G/14 and a text embedding transformer are trained to separately embed images and texts, such that a binary classifier using the sigmoid cross-entropy of the dot product of image and text embeddings correctly classifies whether the respective image and text correspond to each other or not. This is similar to CLIP (Radford et al., 2021) and ALIGN (Jia et al., 2021), but was shown to be more efficient, scalable, and robust (Zhai et al., 2023). This is done in order to pretrain the ViT image embedding component, hence the text embedding transformer is discarded when inserting the ViT into PaLI.

Full PaLI model The outputs of the ViT image encoder before pooling form the visual tokens, which are linearly projected and prepended to the embedded input text tokens. Together, these tokens are passed into a pretrained 3B parameter UL2 encoder-decoder language model (Tay et al., 2023), which generates text output. The text input to the model typically consists of a prompt that describes the type of task (e.g., "*Generate the alt_text in* $\langle \text{lang} \rangle$ *at* $\langle \text{pos} \rangle$" for captioning tasks) and encode necessary textual input for the task (e.g., "*Answer in* $\langle \text{lang} \rangle$*: {question}* " for VQA tasks).

3.2 Stages of Training

The training procedure is similar to that of PaLI and PaLI-X and consists of multiple stages:

Stage 0: Unimodal pretraining. The image encoder is pretrained contrastively on image-text pairs from the web, following the SigLIP training protocol (Zhai et al., 2023). This differs from PaLI and PaLI-X, where a JFT classification pretrained encoder was used. We use a model-based filtering approach similar to that in Schuhmann et al. (2021) that retains about 40% of the pairs. The image encoder is trained at resolution 224×224. The text encoder-decoder is a 3B UL2 model trained following the mixture of denoisers procedure described by Tay et al. (2023).

Stage 1: Multimodal training. Here, the image encoder is combined with the text encoder-decoder as described earlier and in Figure 1. Then, this combined PaLI model is trained on a multimodal task and data mixture, albeit keeping the image encoder `frozen` and using its native

[2]The embedding dimension was changed from 1664 to 1536 for better hardware utilization.

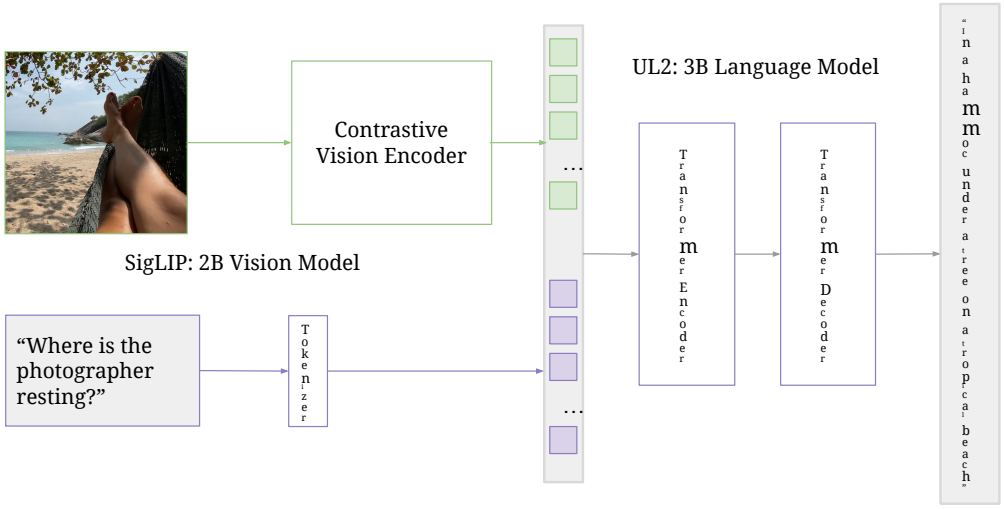


图 1：PaLI-3（5B）模型概览：图像由经过对比预训练的 2B SigLIP 视觉模型分别编码为视觉标记。这些视觉标记连同查询一起传入一个 3B 编码器-解码器 UL2 Transformer，从而生成所需答案。在这种设置下，与先前 PaLI 模型中使用的分类预训练模型相比，经过对比预训练的模型能够提供显著更有用的标记。

视觉组件 PaLI-3 的视觉骨干网络使用 SigLIP（Zhai et al., 2023）训练方案，从一个经过对比预训练的 ViT-G/14[2] (Zhai et al., 2022a) 模型（约 20 亿参数）初始化。简而言之，训练一个图像嵌入 ViT-G/14 和一个文本嵌入 Transformer，使其分别对图像和文本进行嵌入，从而让一个使用图像与文本嵌入点积的 sigmoid 交叉熵的二元分类器，能够正确判断相应的图像和文本是否彼此对应。这类似于 CLIP（Radford et al., 2021）和 ALIGN（Jia et al., 2021），但已被证明更高效、更具可扩展性且更鲁棒（Zhai et al., 2023）。这样做是为了预训练 ViT 图像嵌入组件，因此在将 ViT 插入 PaLI 时会丢弃文本嵌入 Transformer。

完整的 PaLI 模型 ViT 图像编码器在池化之前的输出形成视觉 tokens，这些 tokens 经过线性投影后被前置到嵌入的输入文本 tokens 之前。这些 tokens 一起被输入到一个预训练的 30 亿参数 UL2 编码器-解码器语言模型（Tay 等，2023）中，该模型生成文本输出。模型的文本输入通常由一个描述任务类型的提示组成（例如，用于图像描述任务的“*Generate the alt_text in* ⟨lang⟩ *at* ⟨pos⟩”），并编码该任务所需的文本输入（例如，用于 VQA 任务的“*Answer in* ⟨lang⟩*: {question}*”）。

3.2 训练阶段

训练流程与 PaLI 和 PaLI-X 类似，由多个阶段组成：

阶段 0：单模态预训练。图像编码器按照 SigLIP 训练协议（Zhai 等，2023），在来自网络的图像-文本对上进行对比式预训练。这不同于 PaLI 和 PaLI-X，后两者使用的是经过 JFT 分类预训练的编码器。我们采用了类似于 Schuhmann 等（2021）中的基于模型的过滤方法，保留了约 40% 的配对。图像编码器以 224×224 的分辨率进行训练。文本编码器-解码器是一个 3B UL2 模型，按照 Tay 等（2023）描述的去噪器混合流程进行训练。

阶段 1：多模态训练。在此阶段，图像编码器按照前文及图 1 所述与文本编码器-解码器结合。随后，这个组合后的 PaLI 模型在多模态任务和数据混合上进行训练，但会保持图像编码器冻结并使用其原生

[2]为了更好地利用硬件，嵌入维度从1664改为1536。

Table 1: Performance comparison between contrastively pre-trained ("SigLIP") models and classification pre-trained ("Classif") ViT image encoders using the same PaLI setup, across a wide range of tasks. While linear classification few-shot probing (first column) suggests SigLIP encoders are worse across many tasks, when plugged into PaLI and transferred, they show clear improvements. On the most complicated and detailed image understanding tasks, SigLIP models outperform Classif models by a large margin. Captioning numbers are CIDEr scores, where XM3600 shows the English performance in the first column, and the average across other languages in the second column. RefCOCO numbers are mIoU scores (details in Section 4.3).

		Probe	**Captioning**			**VQA**			**RefCOCO**		
		8 tasks	COCO	XM3600		v2	OK	Text	val	+	g
G/14	Classif	88.1	139.9	94.5	44.7	76.7	57.2	31.9	51.6	43.5	43.4
	SigLIP	**-2.5**	**+0.4**	**+1.6**	**+0.7**	**+0.8**	**+1.4**	**+18.7**	**+15.1**	**+19.1**	**+17.7**
L/16	Classif	86.2	132.6	93.0	42.3	73.7	55.6	24.9	46.9	38.8	38.8
	SigLIP	**-2.8**	**+3.2**	**+1.4**	**+1.4**	**+1.9**	**+1.9**	**+16.2**	**+17.4**	**+20.9**	**+20.1**
B/16	Classif	83.7	127.7	91.7	40.7	72.3	54.7	22.5	46.3	38.1	38.4
	SigLIP	**-2.6**	**+3.6**	**-2.0**	**-0.2**	**+1.4**	**+0.9**	**+13.3**	**+16.8**	**+19.6**	**+19.3**

(224×224) resolution. The main mixture component is again derived from the WebLI dataset by heuristic filtering of the text quality and using the SplitCap training objective (Chen et al., 2023b). Further ingredients inherited from (Chen et al., 2023b) are multilingual captioning on CC3M-35L and WebLI OCR, cross-lingual VQA and VQG using VQ^{2}A-CC3M-35L, object-aware VQA, as well as object detection. Notably, we do not include task or data derived from video (this was done in PaLI-X), but PaLI-3 retains competitive performance on these benchmarks thanks to its strong image encoder. We do, however, further improve document and text understanding capabilities by enriching WebLI with PDF documents with dense text and web-images described as posters or documents, in over 100 languages.

Stage 2: Resolution increase. High-resolution input is a widely accepted way of increasing performance, both due to making more details from the image perceptible, and due to increasing model power via increased sequence length. We increase PaLI-3's resolution by fine-tuning the whole model (unfreezing the image encoder) with a short curriculum of increasing resolutions, keeping checkpoints at 812×812 and 1064×1064 resolution. The data mixture focuses on the part that involves visually-situated text and object detection.

Task specialization (transfer). Finally, for each individual task (benchmark), we fine-tune the PaLI-3 model with frozen ViT image encoder on the task's training data as described in the corresponding section. For most tasks, we fine-tune the 812×812 resolution checkpoint, but for two document understanding tasks, we go up to 1064×1064 resolution.

4 Experiments

4.1 Classification or Contrastively pretrained ViT?

We first perform a controlled comparison of different ViT models within the PaLI framework. We consider two types of ViT models: classification pretrained ("Classif") on the JFT dataset and contrastively pretrained on the WebLI dataset ("SigLIP"). We perform these experiments using a fixed 224×224 resolution (i.e. only include Stage 1) to save compute. We further shorten the Stage 1 phase to 20% of the full PaLI-3 schedule used in the remainder of this paper.

The results in Table 1 paint a clear picture overall: While the few-shot linear classification (Dosovitskiy et al., 2021) of SigLIP models falls behind, when used in PaLI-3, SigLIP models provide moderate gains on "simpler" tasks such as captioning and question-answering, and large gains for more "complicated" scene-text and spatial understanding tasks such as TextVQA and RefCOCO

表 1：在相同的 PaLI 设置下，对比预训练（“SigLIP”）模型与分类预训练（“Classif”）ViT 图像编码器在广泛任务上的性能比较。虽然线性分类少样本探测（第一列）表明 SigLIP 编码器在许多任务上表现更差，但当它们接入 PaLI 并进行迁移时，显示出明显提升。在最复杂且细粒度的图像理解任务上，SigLIP 模型大幅优于 Classif 模型。图像描述生成的数值为 CIDEr 分数，其中 XM3600 第一列显示英语性能，第二列显示其他语言的平均性能。RefCOCO 数值为 mIoU 分数（详见第 4.3 节）。

		Probe	**Captioning**			**VQA**			**RefCOCO**		
		8 tasks	COCO	XM3600		v2	OK	Text	val	+	g
4l/G	Classif	88.1	139.9	94.5	44.7	76.7	57.2	31.9	51.6	43.5	43.4
	SigLIP	**-2.5**	**+0.4**	**+1.6**	**+0.7**	**+0.8**	**+1.4**	**+18.7**	**+15.1**	**+19.1**	**+17.7**
6l/L	Classif	86.2	132.6	93.0	42.3	73.7	55.6	24.9	46.9	38.8	38.8
	SigLIP	**-2.8**	**+3.2**	**+1.4**	**+1.4**	**+1.9**	**+1.9**	**+16.2**	**+17.4**	**+20.9**	**+20.1**
6l/B	Classif	83.7	127.7	91.7	40.7	72.3	54.7	22.5	46.3	38.1	38.4
	SigLIP	**-2.6**	**+3.6**	**-2.0**	**-0.2**	**+1.4**	**+0.9**	**+13.3**	**+16.8**	**+19.6**	**+19.3**

(224×224) 分辨率。主要混合组件同样源自 WebLI 数据集，通过对文本质量进行启发式过滤，并使用 SplitCap 训练目标（Chen et al., 2023b）。继承自（Chen et al., 2023b）的其他组成部分包括 CC3M-35L 和 WebLI OCR 上的多语言图像描述生成、使用 VQ^2A-CC3M-35L 的跨语言 VQA 和 VQG、目标感知 VQA，以及目标检测。值得注意的是，我们没有纳入源自视频的任务或数据（这在 PaLI-X 中已完成），但得益于其强大的图像编码器，PaLI-3 在这些基准上仍保持了有竞争力的性能。不过，我们通过用包含密集文本的 PDF 文档以及被描述为海报或文档的网页图像来丰富 WebLI，进一步提升了文档和文本理解能力，覆盖超过 100 种语言。

阶段 2：提高分辨率。高分辨率输入是一种被广泛接受的提升性能的方法，这既是因为它能让图像中的更多细节变得可感知，也是因为通过增加序列长度提升了模型能力。我们通过使用逐步提高分辨率的短课程对整个模型进行微调（解冻图像编码器），来提高 PaLI-3 的分辨率，并在 812×812 和 1064×1064 分辨率下保留检查点。数据混合侧重于涉及视觉情境文本和目标检测的部分。

任务专门化（迁移）。最后，对于每个单独任务（基准），我们按照相应章节中的描述，在该任务的训练数据上对冻结了 ViT 图像编码器的 PaLI-3 模型进行微调。对于大多数任务，我们微调 812×812 分辨率的检查点，但对于两个文档理解任务，我们将分辨率提高到 1064×1064。

4 实验

4.1 分类预训练还是对比式预训练的 ViT?

我们首先在 PaLI 框架内对不同的 ViT 模型进行受控比较。我们考虑两种类型的 ViT 模型：在 JFT 数据集上进行分类预训练（“Classif”）的模型，以及在 WebLI 数据集上进行对比预训练（“SigLIP”）的模型。为节省计算量，我们使用固定的 224×224 分辨率（即仅包含阶段 1）进行这些实验。我们还将阶段 1 缩短为本文其余部分所用完整 PaLI-3 训练计划的 20%。

表 1 的结果总体上呈现出清晰的图景：尽管 SigLIP 模型的少样本线性分类（Doso- vitskiy 等，2021）表现落后，但当用于 PaLI-3 时，SigLIP 模型在图像描述和问答等“较简单”任务上带来了中等幅度的提升，并在 TextVQA 和 RefCOCO 等更“复杂”的场景文本与空间理解任务上带来了显著提升。

Table 2: Results on benchmarks more focused on understanding visually-situated text. TextCaps, TextVQA, STVQA, InfographicVQA and DocVQA are all evaluated using the corresponding official evaluation server. Methods marked by "†" are trained on additional VQA data similar to the target benchmark, before finetuning on the target benchmark. For ChartQA, we compare with similar setups by finetuning without chain-of-thought or similar prompting techniques. The SOTA models are (a) Chen et al. (2023a), (b) Powalski et al. (2021), (c) Peng et al. (2022).

Model	Text Caps	Text VQA	ST VQA	OCR VQA	Info VQA	Doc VQA	AI2D	Chart QA	Screen2 Words	Widget Cap	Avg. of first 8
***with** OCR pipeline input*											
SOTA	163.7 (a)	**80.78** (a)	84.5 (a)	77.3 (a)	61.2 (b)†	88.4 (c)†	**81.4** (a)	**72.3** (a)	-	-	88.7
PaLI-3	**164.3**	**80.78**	**85.7**	**77.8**	**62.4**	**88.6**	75.2	69.5	-	-	88.0 (-0.7)
***without** OCR pipeline input*											
SOTA	147.0 (a)	71.44 (a)	79.9 (a)	75.0 (a)	49.2 (a)	80.0 (a)	**81.2** (a)	**70.9** (a)	127.9 (a)	153.0 (a)	81.8
PaLI-3	**158.8**	**79.51**	**84.1**	**76.7**	**57.8**	**87.6**	75.2	70.0	**130.7**	**159.8**	86.2 (+4.4)

variants. This motivates the departure from classification pretrained image encoders, and switching to sigmoid-contrastively pretrained ones for building PaLI-3.

4.2 Visually-situated Text Understanding

We evaluate PaLI-3 on visually-situated text understanding tasks: TextCaps (Sidorov et al., 2020), TextVQA (Singh et al., 2019), STVQA (Biten et al., 2019), OCRVQA (Mishra et al., 2019), InfographicVQA (Mathew et al., 2022), DocVQA (Mathew et al., 2021), ChartQA (Masry et al., 2022), Scree2Words (Wang et al., 2021a), and WidgetCap (Li et al., 2020). The images in those datasets span a wide range of domains such as natural images, illustrations, documents and user interfaces.

For the InfographicVQA and DocVQA benchmarks we fine-tune the 1064×1064 resolution model, all others use the 812×812 one. We report the standard metrics for each benchmark, namely: CIDEr score for all the Image Captioning benchmarks; VQA accuracy for VQAv2, OKVQA, and TextVQA; Average Normalized Levenshtein Similarity (ANLS) for ST-VQA, DocVQA and InfographicsVQA; Exact match (EM) for TallyQA, OCR-VQA and AI2D; Relaxed-Accuracy (RA) for ChartQA. For visually-situated text understanding, external OCR systems are usually leveraged to provide OCR annotations of the image as and additional input to the model for boosting performance. Here we follow (Chen et al., 2023a) and report the result of finetuning PaLI-3 both with and without OCR inputs. The OCR annotations are obtained using the same service as that for the training set. As shown in Table 2, PaLI-3 achieves SOTA performance on a vast majority of the captioning and VQA benchmarks both with and without external OCR input. The exception is AI2D and ChartQA, which require not just understanding but also strong reasoning capability over diagrams and charts, respectively. For these two benchmarks, PaLI-3 falls slightly behind PaLI-X (Chen et al., 2023a) likely due to the latter's significantly larger 32B LLM being better at reasoning.

Averaging over the 8 benchmarks in Table 2 that have results in all settings, PaLI-3 is only 0.7 points behind all SOTA methods combined in the setting where external OCR systems are used. However, in the setting without such external system, PaLI-3 has a significant 4.4 point advantage over all SOTA methods combined. For TextCaps, TextVQA, InfographicVQA and DocVQA this advantage is 8 points or more. Finally, we can see that PaLI-3 without any external OCR system is only 1.8 points behind relying on one, suggesting the image encoder learns a strong intrinsic OCR capability.

4.3 Referring Expression Segmentation

We extend PaLI-3 with the capability to predict segmentation masks via language-like output. To this end, we make use of the vector-quantized variational auto-encoder (VQ-VAE) from Ning et al. (2023). The VQ-VAE is trained to learn a discrete codebook of 128 mask tokens. Its encoder can

表 2：在更侧重于理解视觉场景中文本的基准上的结果。TextCaps、TextVQA、STVQA、InfographicVQA 和 DocVQA 均使用相应的官方评测服务器进行评估。标记为“†”的方法在目标基准上微调之前，使用了与目标基准类似的额外 VQA 数据进行训练。对于 ChartQA，我们通过在不使用思维链或类似提示技术的情况下进行微调，与类似设置进行比较。SOTA 模型包括 (a) Chen 等人（2023a），(b) Powalski 等人（2021），(c) Peng 等人（2022）。

Model	Text Caps	Text VQA	ST VQA	OCR VQA	Info VQA	Doc VQA	AI2D	Chart QA	Screen2 Words	Widget Cap	Avg. of first 8
***with** OCR pipeline input*											
SOTA	163.7 (a)	**80.78** (a)	84.5 (a)	77.3 (a)	61.2 (b)†	88.4 (c)†	**81.4** (a)	**72.3** (a)	-	-	88.7
PaLI-3	**164.3**	**80.78**	**85.7**	**77.8**	**62.4**	**88.6**	75.2	69.5	-	-	88.0 (-0.7)
***without** OCR pipeline input*											
SOTA	147.0 (a)	71.44 (a)	79.9 (a)	75.0 (a)	49.2 (a)	80.0 (a)	**81.2** (a)	**70.9** (a)	127.9 (a)	153.0 (a)	81.8
PaLI-3	**158.8**	**79.51**	**84.1**	**76.7**	**57.8**	**87.6**	75.2	70.0	**130.7**	**159.8**	86.2 (+4.4)

变体。这促使我们不再采用经过分类预训练的图像编码器，转而改用经过 sigmoid 对比式预训练的图像编码器来构建 PaLI-3。

4.2 视觉情境下的文本理解

我们在视觉情境文本理解任务上评估 PaLI-3：TextCaps（Sidorov 等，2020）、TextVQA（Singh 等，2019）、STVQA（Biten 等，2019）、OCRVQA（Mishra 等，2019）、Info- graphicVQA（Mathew 等，2022）、DocVQA（Mathew 等，2021）、ChartQA（Masry 等，2022）、Scree2Words（Wang 等，2021a）和 WidgetCap（Li 等，2020）。这些数据集中的图像涵盖了广泛的领域，例如自然图像、插图、文档和用户界面。

对于 InfographicVQA 和 DocVQA 基准，我们微调 1064×1064 分辨率模型，其他所有基准均使用 812×812 分辨率模型。我们报告每个基准的标准指标，即：所有图像描述基准使用 CIDEr 分数；VQAv2、OKVQA 和 TextVQA 使用 VQA 准确率；ST-VQA、DocVQA 和 InfographicsVQA 使用平均归一化 Levenshtein 相似度（ANLS）；TallyQA、OCR-VQA 和 AI2D 使用精确匹配（EM）；ChartQA 使用宽松准确率（RA）。对于视觉场景中的文本理解，通常会利用外部 OCR 系统提供图像的 OCR 标注，作为模型的额外输入以提升性能。这里我们遵循（Chen et al., 2023a），并报告在有无 OCR 输入两种情况下微调 PaLI-3 的结果。OCR 标注使用与训练集相同的服务获得。如表 2 所示，无论是否使用外部 OCR 输入，PaLI-3 在绝大多数图像描述和 VQA 基准上都达到了 SOTA 性能。例外是 AI2D 和 ChartQA，它们分别要求模型不仅理解图表和图形，还需要具备强大的推理能力。在这两个基准上，PaLI-3 略逊于 PaLI-X（Chen et al., 2023a），这可能是因为后者显著更大的 32B LLM 更擅长推理。

对表 2 中在所有设置下都有结果的 8 个基准进行平均后，PaLI-3 在使用外部 OCR 系统的设置下，仅比所有 SOTA 方法的综合结果低 0.7 分。然而，在没有此类外部系统的设置下，PaLI-3 相比所有 SOTA 方法的综合结果具有显著的 4.4 分优势。对于 TextCaps、TextVQA、InfographicVQA 和 DocVQA，这一优势达到 8 分或更多。最后，我们可以看到，不使用任何外部 OCR 系统的 PaLI-3 仅比依赖外部 OCR 系统低 1.8 分，这表明图像编码器学习到了强大的内在 OCR 能力。

4.3 指代表达式分割

我们扩展了 PaLI-3，使其具备通过类似语言的输出预测分割掩码的能力。为此，我们使用了 Ning 等人（2023）的向量量化变分自编码器（VQ-VAE）。该 VQ-VAE 经过训练，用于学习一个包含 128 个掩码词元的离散码本。其编码器可以

Table 3: PaLI referring expression segmentation results on RefCOCO (Yu et al., 2016) variants. All results are mIoU on the `val` split.

Model	RefCOCO	RefCOCO+	G-Ref
RefTr (Li & Sigal, 2021)	74.34	66.75	66.63
PolyFormer (Liu et al., 2023)	76.94	72.15	71.15
PaLI-3 (Ours)	**77.33**	**73.53**	**72.72**

tokenize a 64×64 pixels segmentation mask into 16 mask tokens, which its decoder can convert back. We train PaLI-3 to predict a single segmentation mask. First, PaLI-3 outputs 4 coordinates as text, representing a bounding box. This is followed by 16 mask tokens that represent the mask inside the bounding box.

We fine-tune PaLI-3 on the combined training sets of RefCOCO, RefCOCO+, and RefCOCOg (Yu et al., 2016)[3] at 812×812 resolution. Each training example consists of a referring expression (e.g. "the large brown dog on the left"), and a box with segmentation mask. We prompt PaLI with the prompt "*detect: the large brown dog on the left* $\langle \text{extra_id_0} \rangle$", and the target is a sequence like "*348 543 684 664* $\langle \text{mask_token_81} \rangle \dots \langle \text{mask_token_10} \rangle$". The target sequence contains 16 mask tokens between 0 and 127 that are generated by the VQ-VAE encoder using the segmentation mask cropped and resized to 64×64 as input.

The results in Table 1 demonstrate that contrastive pretraining is much more effective than classification pretraining for localization task of this type. Table 3 shows that the full PaLI-3 model is able to slightly outperform the state of the art in referring expression segmentation.

4.4 Natural Image Understanding

In this section, we evaluate PaLI-3 on general vision-language understanding tasks, including COCO captions (Karpathy & Fei-Fei, 2015) and VQAv2 (Goyal et al., 2017) which target general visual understanding, OKVQA (Marino et al., 2019) which focuses on knowledge-based understanding and TallyQA (Acharya et al., 2019) which measures performance on counting under textual guidance. All results for the benchmarks presented in Table 4 use 812×812 resolution. As in previous work, they employ no external OCR module, since these benchmarks rarely involve text in the image.

Overall, PaLI-3 shows very strong performance on these benchmarks despite its significantly smaller size compared to recent SOTA models. For COCO, PaLI-3 outperforms all models but BEiT-3 and the 17B and 55B PaLI. On VQAv2 and TallyQA, PaLI-3 exceeds all previous models except PaLI-X, with a less than 1 point gap on VQAv2. For the knowledge-intensive OKVQA task, which usually benefits from a large language component, PaLI-3 is only behind PaLM-E (562B) and PaLI-X (55B) but still outperforms the 32-shot Flamingo (80B) model.

4.5 Video Captioning and Question Answering

We fine-tune and evaluate the PaLI-3 model on 4 video captioning benchmarks: MSR-VTT (Xu et al., 2016), VATEX (Wang et al., 2019), ActivityNet Captions (Krishna et al., 2017), and Spoken Moments in Time (Monfort et al., 2021). We do the same for 3 video question-answering benchmarks: NExT-QA (Xiao et al., 2021), MSR-VTT-QA (Xu et al., 2017), and ActivityNet-QA (Yu et al., 2019). A brief description of each benchmark and its usage is provided in Appendix A.

Following the setup from PaLI-X (Chen et al., 2023a), we fine-tune our model using the Stage 1 checkpoint with 224×224 resolution for each task separately. We sample at most 16 frames with a fixed temporal stride for each benchmark. Each frame is independently processed by the ViT image encoder, the resulting visual tokens are simply concatenated, leading to up to 4096 visual tokens. A key difference from the PaLI-X setup is that there is no video data in PaLI-3 pretraining, meaning PaLI-3 has never seen multi-frame inputs during pretraining.

[3]We removed all validation and test images from the training set for both PaLI and the VQ-VAE

表 3：PaLI 在 RefCOCO（Yu 等，2016）变体上的指代表达式分割结果。所有结果均为验证集上的 mIoU。

Model	RefCOCO	RefCOCO+	G-Ref
RefTr (Li & Sigal, 2021)	74.34	66.75	66.63
PolyFormer (Liu et al., 2023)	76.94	72.15	71.15
PaLI-3 (Ours)	**77.33**	**73.53**	**72.72**

将一个 64 × 64 像素的分割掩码标记化为 16 个掩码标记，其解码器可以将其转换回来。我们训练 PaLI-3 来预测单个分割掩码。首先，PaLI-3 以文本形式输出 4 个坐标，表示一个边界框。随后输出 16 个掩码标记，表示边界框内的掩码。

我们在 RefCOCO、RefCOCO+ 和 RefCOCOg（Yu 等，2016）[3] 的组合训练集上，以 812×812 分辨率对 PaLI-3 进行微调。每个训练样本由一个指代表达（例如“左边那只棕色的大狗”）和一个带分割掩码的框组成。我们使用提示“*detect: the large brown dog on the left* ⟨extra_id_0⟩”来提示 PaLI，目标是类似“*348 543 684 664* ⟨mask_token_81⟩ . . . ⟨mask_token_10⟩”的序列。目标序列包含 16 个介于 0 到 127 之间的掩码标记，这些标记由 VQ-VAE 编码器生成，其输入是裁剪并调整大小到 64 × 64 的分割掩码。

表 1 的结果表明，对于此类定位任务，对比预训练比分类预训练有效得多。表 3 显示，完整的 PaLI-3 模型在指代表达式分割任务上能够略微优于当前最先进水平。

4.4 自然图像理解

在本节中，我们在通用视觉-语言理解任务上评估 PaLI-3，包括面向通用视觉理解的 COCO captions（Karpathy & Fei-Fei, 2015）和 VQAv2（Goyal et al., 2017）、侧重于基于知识的理解的 OKVQA（Marino et al., 2019），以及衡量在文本指导下进行计数表现的 TallyQA（Acharya et al., 2019）。表 4 中给出的各项基准测试结果均使用 812×812 分辨率。与以往工作一样，它们未采用外部 OCR 模块，因为这些基准测试很少涉及图像中的文本。

总体而言，尽管 PaLI-3 的规模明显小于近期的 SOTA 模型，但它在这些基准测试上表现非常强劲。在 COCO 上，PaLI-3 的表现优于除 BEiT-3 以及 17B 和 55B PaLI 之外的所有模型。在 VQAv2 和 TallyQA 上，PaLI-3 超过了除 PaLI-X 之外的所有先前模型，并且在 VQAv2 上的差距不到 1 分。对于通常受益于大型语言组件的知识密集型 OKVQA 任务，PaLI-3 仅落后于 PaLM-E（562B）和 PaLI-X（55B），但仍优于 32-shot Flamingo（80B）模型。

4.5 视频字幕生成与问答

我们在 4 个视频字幕生成基准上微调并评估 PaLI-3 模型：MSR-VTT（Xu 等，2016）、VATEX（Wang 等，2019）、ActivityNet Captions（Krishna 等，2017）以及 Spoken Moments in Time（Monfort 等，2021）。我们也在 3 个视频问答基准上进行了同样的操作：NExT-QA（Xiao 等，2021）、MSR-VTT-QA（Xu 等，2017）以及 ActivityNet-QA（Yu 等，2019）。附录 A 中提供了每个基准及其用法的简要说明。

沿用 PaLI-X（Chen et al., 2023a）的设置，我们使用 Stage 1 检查点，以 224×224 分辨率分别针对每个任务对模型进行微调。对于每个基准，我们以固定的时间步幅最多采样 16 帧。每一帧都由 ViT 图像编码器独立处理，得到的视觉 token 被简单拼接，最多可产生 4096 个视觉 token。与 PaLI-X 设置的一个关键区别在于，PaLI-3 的预训练中没有视频数据，这意味着 PaLI-3 在预训练期间从未见过多帧输入。

[3] 我们从PaLI和VQ-VAE的训练集中移除了所有验证图像和测试图像。

Table 4: Results on COCO Captions (Karpathy split), VQAv2, OKVQA, and TallyQA. (*Flamingo reports 32 shot result). Underscored numbers indicate that PaLI-3 is only behind the 55B PaLI-X and is better than all other models in the list.

	COCO	VQAv2		OKVQA	TallyQA	
Model	Karp.-test	test-dev	test-std	val	Simple	Complex
SimVLM	143.3	80.03	80.34	-	-	-
CoCa (2.1B)	143.6	82.3	82.3	-	-	-
GIT (0.7B)	144.8	78.56	78.81	-	-	-
GIT2 (5.1B)	145.0	81.74	81.92	-	-	-
OFA (0.9B)	145.3	82.0	82.0	-	-	-
Flamingo (80B)	138.1	82.0	82.1	57.8*	-	-
BEiT-3 (1.9B)	147.6	84.2	84.0	-	-	-
PaLM-E (562B)	138.7	80.0	-	**66.1**	-	-
MoVie	-	69.26	-	-	74.9	56.8
PaLI-17B	149.1	84.3	84.3	64.5	81.7	70.9
PaLI-X (55B)	**149.2**	**86.0**	**86.1**	**66.1**	**86.0**	**75.6**
PaLI-3 (5B)	145.9	<u>85.0</u>	<u>85.2</u>	60.1	<u>83.3</u>	70.5

Table 5: Results for Video Captioning and Video-QA using up to 16 frames. †GIT2 directly optimizes the CIDEr metric. mPLUG-2 is Xu et al. (2023), PaLI-X is Chen et al. (2023a), GIT2 is Wang et al. (2022a), and Flamingo-32 is the 32-shot variant of Alayrac et al. (2022).

	MSR-VTT		Activity-Net		VATEX	SMIT	NExT-QA
Method	Caption	QA	Caption	QA	Caption	Caption	QA
Prior SOTA	**80.3** mPLUG-2	48.0 mPLUG-2	**54.9** PaLI-X	49.4 PaLI-X	94.0† GIT2	**43.5** PaLI-X	**38.3** Flamingo-32
PaLI-3	78.3	**49.3**	50.8	**51.2**	66.9	39.6	37.7

Despite not being pretrained with video data, PaLI-3 achieves excellent video QA results with a small model size: a new state of the art performance on MSR-VTT-QA and ActivityNet-QA, and competitive results on NextQA. The consistent improvements on image and video QA highlight the benefits of adopting the contrastive ViTs. PaLI-3 also achieves respectable video captioning results, under-performing the SOTA by only 3 CIDEr points on average. Considering the model size, PaLI-3 appears to be an excellent choice in terms of both performance and practicality.

Table 6: Evaluations of the visual component in isolation (without the language model). We report fine-tuned classification accuracy on ImageNet, ImageNet-ReaL and ImageNet-V2; average zero-shot cross-modal retrieval recall@1 across 36 languages on XM3600; average 10-shot linear probe classification accuracy across 8 tasks.

Model	**Encoder**	**ImageNet (fine-tuning)**				**XM3600 (retrieval)**		**Probe**
		Res.	Val	ReaL	v2	I→T	T→I	8 tasks
PaLI-3	SigLIP ViT-G	518px	89.6	90.9	82.3	56.9	44.0	85.6
PaLI-15B	Classif ViT-G	518px	90.5	90.8	83.3	-	-	88.1
PaLI-17B	Classif ViT-e	644px	90.9	91.1	84.3	36.0	28.5	89.5
PaLI-X	Classif ViT-22B	756px	89.2	91.0	83.7	-	-	89.9

表 4：在 COCO Captions（Karpathy 划分）、VQAv2、OKVQA 和 TallyQA 上的结果。（*Flamingo 报告的是 32-shot 结果）。带下划线的数字表示 PaLI-3 仅次于 55B PaLI-X，并且优于列表中的所有其他模型。

	COCO	VQAv2		OKVQA	TallyQA	
Model	Karp.-test	test-dev	test-std	val	Simple	Complex
SimVLM	143.3	80.03	80.34	-	-	-
CoCa (2.1B)	143.6	82.3	82.3	-	-	-
GIT (0.7B)	144.8	78.56	78.81	-	-	-
GIT2 (5.1B)	145.0	81.74	81.92	-	-	-
OFA (0.9B)	145.3	82.0	82.0	-	-	-
Flamingo (80B)	138.1	82.0	82.1	57.8*	-	-
BEiT-3 (1.9B)	147.6	84.2	84.0	-	-	-
PaLM-E (562B)	138.7	80.0	-	**66.1**	-	-
MoVie	-	69.26	-	-	74.9	56.8
PaLI-17B	149.1	84.3	84.3	64.5	81.7	70.9
PaLI-X (55B)	**149.2**	**86.0**	**86.1**	**66.1**	**86.0**	**75.6**
PaLI-3 (5B)	145.9	<u>85.0</u>	<u>85.2</u>	60.1	<u>83.3</u>	70.5

表 5：使用最多 16 帧进行视频字幕生成和视频问答的结果。†GIT2 直接优化 CIDEr 指标。mPLUG-2 为 Xu 等人（2023），PaLI-X 为 Chen 等人（2023a），GIT2 为 Wang 等人（2022a），Flamingo-32 为 Alayrac 等人（2022）的 32-shot 变体。

	MSR-VTT		Activity-Net		VATEX	SMIT	NExT-QA
Method	Caption	QA	Caption	QA	Caption	Caption	QA
Prior SOTA	**80.3** mPLUG-2	48.0 mPLUG-2	**54.9** PaLI-X	49.4 PaLI-X	94.0† GIT2	**43.5** PaLI-X	**38.3** Flamingo-32
PaLI-3	78.3	**49.3**	50.8	**51.2**	66.9	39.6	37.7

尽管未使用视频数据进行预训练，PaLI-3 仍以较小的模型规模取得了出色的视频问答结果：在 MSR-VTT-QA 和 ActivityNet-QA 上达到新的最先进性能，并在 NextQA 上取得具有竞争力的结果。图像和视频问答上的持续提升凸显了采用对比式 ViT 的优势。PaLI-3 还取得了相当不错的视频字幕生成结果，平均仅比 SOTA 低 3 个 CIDEr 分。考虑到模型规模，PaLI-3 在性能和实用性方面似乎都是一个出色的选择。

表 6：对视觉组件进行单独评估（不包含语言模型）。我们报告了在 ImageNet、ImageNet-ReaL 和 ImageNet-V2 上的微调分类准确率；在 XM3600 上跨 36 种语言的平均零样本跨模态检索 recall@1；以及跨 8 个任务的平均 10-shot 线性探针分类准确率。

Model	**Encoder**	**ImageNet (fine-tuning)**				**XM3600 (retrieval)**		**Probe**
		Res.	Val	ReaL	v2	I→T	T→I	8 tasks
PaLI-3	SigLIP ViT-G	518px	89.6	90.9	82.3	56.9	44.0	85.6
PaLI-15B	Classif ViT-G	518px	90.5	90.8	83.3	-	-	88.1
PaLI-17B	Classif ViT-e	644px	90.9	91.1	84.3	36.0	28.5	89.5
PaLI-X	Classif ViT-22B	756px	89.2	91.0	83.7	-	-	89.9

Table 7: RAI statistics for captions generated by PaLI-3 on FairFace (Karkkainen & Joo, 2021).

	Perceived Gender		Ethnicity			Age Bucket			
	Lowest	Highest	Lowest	Median	Highest	Lowest	Median	Highest	**Overall**
Toxicity	0.02%	0.05%	0.00%	0.00%	0.10%	0.00%	0.00%	0.07%	**0.04%**
Profanity	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	**0.00%**
Insult	0.04%	0.10%	0.00%	0.07%	0.14%	0.00%	0.06%	0.17%	**0.07%**
Threat	0.10%	0.12%	0.00%	0.14%	0.20%	0.00%	0.00%	0.21%	**0.11%**
Attack	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	**0.00%**

Table 8: RAI score statistics in the captions generated by PaLI-3 on MIAP (Schumann et al., 2021).

	Perceived Gender		Age Bucket			Skin Tone			
	Lowest	Highest	Lowest	Median	Highest	Lowest	Median	Highest	**Overall**
Toxicity	0.05%	0.10%	0.05%	0.24%	0.48%	0.00%	0.00%	0.26%	**0.07%**
Profanity	0.10%	0.10%	0.00%	0.12%	0.24%	0.00%	0.00%	0.10%	**0.10%**
Insult	0.10%	0.14%	0.00%	0.13%	0.17%	0.00%	0.00%	0.38%	**0.13%**
Threat	0.34%	0.80%	0.30%	0.54%	1.68%	0.00%	0.59%	0.94%	**0.65%**
Identity Attack	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	**0.00%**

4.6 Direct image encoder evaluation

Here, we aim to directly evaluate the learned image encoder (ViT-G model) without the surrounding language model, i.e. not the full PaLI-3. All results are summarized in Table 6.

First, we test image classification capabilities using the standard ImageNet (Russakovsky et al., 2014) benchmark and two of its most popular variants (Beyer et al., 2020; Recht et al., 2019). We fine-tune the unimodal image encoder from Stage 0 on ImageNet and compare it to fine-tuning of classification pretrained ViTs used in previous PaLI models. The main comparison is to the classification (Classif) pretrained ViT-G/14 model from Zhai et al. (2022a). The SigLIP slightly lags behind in terms of top-1 and v2 accuracy, but matches in terms of ReaL accuracy (Beyer et al., 2020), a metric which avoids measuring "overfitting" to ImageNet peculiarities.

Second, we report multilingual image-text retrieval results on the Crossmodal-3600 benchmark (Thapliyal et al., 2022). Since classification pretrained image encoders do not have this capability, we LiT-tune (Zhai et al., 2022b) a text encoder for it on the multilingual WebLI dataset. We also LiT-tune a new text encoder for the SigLIP image encoder in the exact same setting, to remove any confounders from the comparison. The SigLIP ViT-G model clearly outperforms the classification pretrained larger ViT-e model.

Finally, we perform linear probing of the representation in the few-shot setting following Dosovitskiy et al. (2021); Zhai et al. (2022a) across the 8 different classification tasks also used in Zhai et al. (2022a) and report the average performance. Here, we see that SigLIP lags behind, likely because the representation is not pretrained in a way that supports linear separability, as was recently uncovered by Tschannen et al. (2023).

Taken together, these results paint a clear picture: the best and largest classification pretrained image encoders appear (slightly) better when evaluated on standard classification tasks, however they are significantly worse than SigLIP pretrained image encoders for vision-language tasks.

5 Model Fairness, Biases, and Other Potential Issues

We follow the evaluation protocol of Chen et al. (2023a) to assess the model fairness, biases, and other potential issues. First, we use the MIAP (Schumann et al., 2021) and FairFace (Karkkainen & Joo, 2021) datasets to generate captions and use the Perspective API (Lees et al., 2022) (threshold > 0.8) to measure toxicity and profanity among other potential issues. Table 7 (FairFace) and Table 8 (MIAP) summarize the results. Slices with < 20 examples are dropped from the analysis. Overall, we observe a low level of toxicity and profanity among others, across all slices. The results are comparable to those in PaLI-X (Chen et al., 2023a).

表 7：PaLI-3 在 FairFace（Karkkainen & Joo，2021）上生成的标题的 RAI 统计数据。

	Perceived Gender		Ethnicity			Age Bucket			
	Lowest	Highest	Lowest	Median	Highest	Lowest	Median	Highest	**Overall**
Toxicity	0.02%	0.05%	0.00%	0.00%	0.10%	0.00%	0.00%	0.07%	**0.04%**
Profanity	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	**0.00%**
Insult	0.04%	0.10%	0.00%	0.07%	0.14%	0.00%	0.06%	0.17%	**0.07%**
Threat	0.10%	0.12%	0.00%	0.14%	0.20%	0.00%	0.00%	0.21%	**0.11%**
Attack	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	**0.00%**

表 8：PaLI-3 在 MIAP（Schumann 等，2021）上生成的说明文字中的 RAI 分数统计。

	Perceived Gender		Age Bucket			Skin Tone			
	Lowest	Highest	Lowest	Median	Highest	Lowest	Median	Highest	**Overall**
Toxicity	0.05%	0.10%	0.05%	0.24%	0.48%	0.00%	0.00%	0.26%	**0.07%**
Profanity	0.10%	0.10%	0.00%	0.12%	0.24%	0.00%	0.00%	0.10%	**0.10%**
Insult	0.10%	0.14%	0.00%	0.13%	0.17%	0.00%	0.00%	0.38%	**0.13%**
Threat	0.34%	0.80%	0.30%	0.54%	1.68%	0.00%	0.59%	0.94%	**0.65%**
Identity Attack	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	**0.00%**

4.6 直接图像编码器评估

在这里，我们旨在直接评估学习到的图像编码器（ViT-G 模型），而不包括外围的语言模型，也就是说不是完整的 PaLI-3。所有结果汇总在表 6 中。

首先，我们使用标准的 ImageNet（Russakovsky 等，2014）基准及其两个最受欢迎的变体（Beyer 等，2020；Recht 等，2019）来测试图像分类能力。我们在 ImageNet 上对阶段 0 的单模态图像编码器进行微调，并将其与先前 PaLI 模型中使用的分类预训练 ViT 的微调结果进行比较。主要比较对象是 Zhai 等（2022a）提出的分类（Classif）预训练 ViT-G/14 模型。SigLIP 在 top-1 和 v2 准确率方面略微落后，但在 ReaL 准确率（Beyer 等，2020）方面持平；该指标可避免衡量对 ImageNet 特有性质的“过拟合”。

其次，我们报告了在 Crossmodal-3600 基准（Thapliyal 等，2022）上的多语言图文检索结果。由于分类预训练的图像编码器不具备这种能力，我们在多语言 WebLI 数据集上为其对文本编码器进行 LiT 微调（Zhai 等，2022b）。我们还在完全相同的设置下为 SigLIP 图像编码器对一个新的文本编码器进行 LiT 微调，以消除比较中的任何混杂因素。SigLIP ViT-G 模型明显优于经过分类预训练且规模更大的 ViT-e 模型。

最后，我们按照 Dosovitskiy 等人（2021）；Zhai 等人（2022a）的方法，在少样本设置下，对表征进行线性探测，涵盖 Zhai 等人（2022a）中也使用的 8 个不同分类任务，并报告平均性能。在这里，我们看到 SigLIP 表现落后，这可能是因为其表征并未以支持线性可分性的方式进行预训练，正如 Tschannen 等人（2023）最近所揭示的那样。

综上所述，这些结果勾勒出一幅清晰的图景：在标准分类任务上评估时，最好且最大的经过分类预训练的图像编码器似乎（略微）更好；然而，在视觉-语言任务上，它们明显不如经过 SigLIP 预训练的图像编码器。

5 模型公平性、偏见和其他潜在问题

我们遵循 Chen 等人（2023a）的评估协议来评估模型公平性、偏见以及其他潜在问题。首先，我们使用 MIAP（Schumann 等人，2021）和 FairFace（Karkkainen & Joo，2021）数据集生成字幕，并使用 Perspective API（Lees 等人，2022）（阈值 > 0.8）来衡量毒性、亵渎性语言以及其他潜在问题。表 7（FairFace）和表 8（MIAP）总结了结果。包含 < 20 个示例的切片会从分析中剔除。总体而言，我们观察到在所有切片中，毒性和亵渎性语言等问题的水平都较低。结果与 PaLI-X（Chen 等人，2023a）中的结果相当。

Table 9: Detection error rate for "person" in PaLI-3 using the subset of the MIAP dataset (Schumann et al., 2021) that contain exactly a single individual in the image. PaLI-3 maintains a low error rate across all subgroups. Skin tone follows the Monk Skin Tone Scale (Monk, 2019). Numbers inside square brackets correspond to the size of each bucket.

Skin Tone	**1** [2]	**2** [871]	**3** [3008]	**4** [522]	**5** [184]	**6** [85]	**7** [54]	**8** [49]	**9** [6]	**10** [1]
	0.00%	0.00%	0.17%	0.39%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%

Gender	**Predominantly Feminine** [2437]	**Predominantly Masculine** [3544]
	0.41%	1.78%

Age Bucket	**0-2 yrs** [17]	**3-19 yrs** [568]	**20-59 yrs** [4925]	**> 60 yrs** [247]
	0.00%	0.18%	1.30%	0.82%

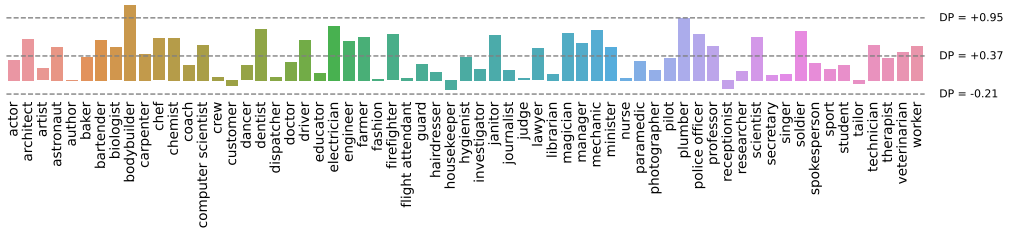


Figure 2: Level of demographic parity (DP) in PaLI-3's output for CelebA images, comparing the average log-perplexity between females and males. Values close to zero indicate absence of bias. The dotted lines correspond to μ and $\mu \pm 2\sigma$. DP for all occupations falls within the 95% confidence interval, except "bodybuilder" and "plumber" which the model tends to strongly associate with men.

Second, we examine the level of demographic parity (Dwork et al., 2012) in the model itself. Following (Chen et al., 2023a), we feed an image from the CelebA dataset (Liu et al., 2015) into PaLI-3 with the chosen occupation as a prefix and record the average log-perplexity score of the model generation. The demographic parity is the difference between average log-perplexity within the demographic groups. Figure 2 summarizes the results: Similar to PaLI-X, PaLI-3 tends to assign a *higher* log-perplexity score to women than men across most occupations with a mean difference of $\mu = 0.37$. However, fewer occupations in PaLI-3 fall outside the interval $\mu \pm 2\sigma$ compared to PaLI-X.

Third, we compare performance across all subgroups on a detection task using the MIAP dataset, following again (Chen et al., 2023a). For images containing exactly a single individual, we query PaLI-3 with the question: "Is there a person in this image?" and evaluate the accuracy of its response. Table 9 summarizes the results. The error rate (false negatives) is very low across all subgroups.

For analysis of the WebLI dataset itself, such as the correlations between perceived gender and occupations, see Chen et al. (2023a).

Limitations. The limitations of this work are very similar to those already presented in the literature. We refer to (Chen et al., 2023a), which raises all limitations that apply to our work.

6 Conclusion

In this paper, we took a closer look at the pretraining of the image encoder in large VLMs, specifically the PaLI type of model. By performing controlled experiments on that part, for the first time, we clearly compare the two camps of classification pretraining and image-text (contrastive) pretraining, finding that the latter can lead to better and more efficient VLMs, especially for localization and text understanding tasks. This is just *one small aspect* of VLMs, and we hope this study and result spurs to further detailed investigations of the *many other aspects* of training VLMs.

表 9：在 MIAP 数据集（Schumann 等，2021）中图像恰好包含单个个体的子集上，PaLI-3 对“person”的检测错误率。PaLI-3 在所有子群体中均保持较低的错误率。肤色遵循 Monk 肤色量表（Monk，2019）。方括号内的数字对应于每个分桶的大小。

Skin Tone	**1** [2]	**2** [871]	**3** [3008]	**4** [522]	**5** [184]	**6** [85]	**7** [54]	**8** [49]	**9** [6]	**10** [1]
	0.00%	0.00%	0.17%	0.39%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
Gender	**Predominantly Feminine** [2437]					**Predominantly Masculine** [3544]				
	0.41%					1.78%				
Age Bucket	**0-2 yrs** [17]		**3-19 yrs** [568]		**20-59 yrs** [4925]		**> 60 yrs** [247]			
	0.00%		0.18%		1.30%		0.82%			

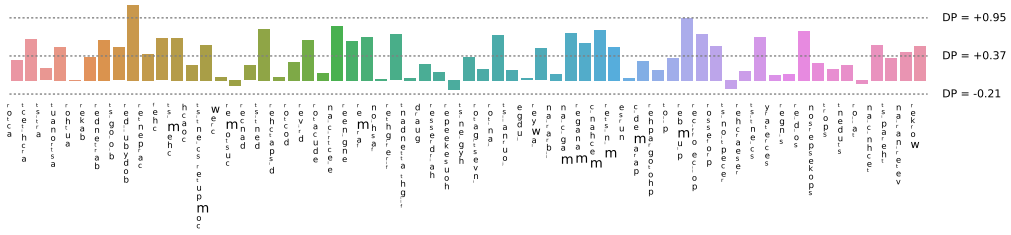


图 2：PaLI-3 对 CelebA 图像输出中的人口统计均等性（DP）水平，比较女性与男性之间的平均对数困惑度。接近零的值表示不存在偏差。虚线对应于 μ 和 $\mu \pm 2\sigma$。除“健美运动员”和“水管工”外，所有职业的 DP 都落在 95% 置信区间内；模型往往将这两个职业强烈地与男性相关联。

其次，我们考察模型本身的人口统计均等性（Dwork et al., 2012）水平。按照（Chen et al., 2023a）的方法，我们将 CelebA 数据集（Liu et al., 2015）中的一张图像输入 PaLI-3，并将所选职业作为前缀，记录模型生成结果的平均对数困惑度分数。人口统计均等性是人口统计群体内平均对数困惑度之间的差异。图 2 总结了结果：与 PaLI-X 类似，在大多数职业中，PaLI-3 倾向于给女性分配比男性*higher*的对数困惑度分数，平均差异为 $\mu = 0.37$。然而，与 PaLI-X 相比，PaLI-3 中落在区间 $\mu \pm 2\sigma$ 之外的职业更少。

第三，我们再次遵循（Chen et al., 2023a），使用 MIAP 数据集比较所有子群体在检测任务上的表现。对于只包含一个人的图像，我们向 PaLI-3 提问：“这张图像中有人吗？”并评估其回答的准确性。表 9 总结了结果。所有子群体的错误率（假阴性）都非常低。

关于 WebLI 数据集本身的分析，例如感知性别与职业之间的相关性，参见 Chen 等人（2023a）。

局限性。本工作的局限性与文献中已提出的局限性非常相似。我们参考（Chen et al., 2023a），其中提出了所有适用于我们工作的局限性。

6 结论

在本文中，我们更深入地研究了大型 VLM 中图像编码器的预训练，具体来说是 PaLI 类型的模型。通过对该部分进行受控实验，我们首次明确比较了分类预训练和图像-文本（对比）预训练这两大阵营，发现后者能够带来更好且更高效的 VLM，尤其是在定位和文本理解任务上。这只是 VLM 的 *one small aspect*，我们希望这项研究及其结果能够推动对训练 VLM 的 *many other aspects* 进行进一步的详细研究。

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7 致谢

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A ADDITIONAL RESULTS: VIDEO CAPTIONING AND QA

A.1 DATASETS & BENCHMARKS

Because not all videos in the benchmarks are available online at the time of experimentation when freshly collected the data, the effective numbers of videos is smaller than the public official splits in some cases. Table 10 reports the details numbers for the subset used in our training and evaluation. We follows the same experimental settings used in PaLI-X (Chen et al., 2023a), including the dataset splits configuration and the evaluation metrics. Please refer to PaLI-X (Chen et al., 2023a) for more data-related details.

		MSR-VTT	VATEX	ANet-Cap	SMIT	M-V-QA	. ANet-QA	NExT-QA
Original size	valid.	497	3000	17505	14604	12278	18000	5343
	test	2990	6000	17031	3513	72821	8000	9178
Dataset size	valid.	325	2646	14566	8096	8160	10000	5343
	test	2135	5242	14197	3513	52623	7040	9178
% Remaining	valid.	65.39	88.20	83.21	100.00	66.46	-	100.00
	test	71.40	87.37	83.36	100.00	72.26	88.00	100.00

Table 10: As we freshly collect the data sets, the actual amount of training data is smaller than the public benchmarks, making the tasks more challenging. Except for NextQA and SMIT, there are more than 10% of the videos missing in both training and evaluation.

B ADDITIONAL RESULTS: CROSSMODAL-3600 RETRIEVAL

We present more results of zero-shot image-text retrieval results (recall@1) on Crossmodal-3600 (Thapliyal et al., 2022) in this section. Detailed results across 36 languages for SigLIP ViT-G and Classif ViT-e are presented in Figure 3, Figure 4 and Table 11.

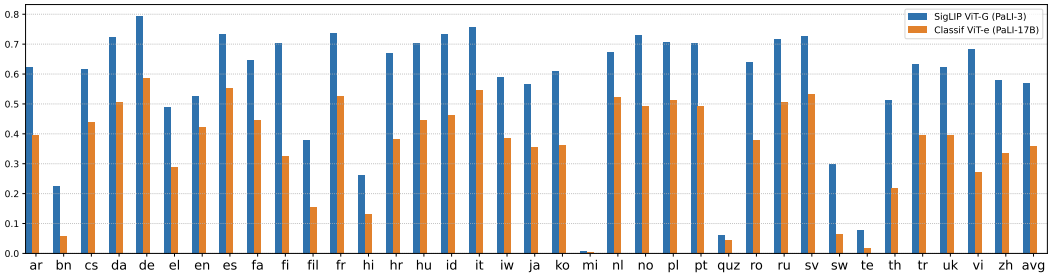


Figure 3: Image-to-text zero-shot retrieval recall@1 on crossmodal-3600 for SigLIP ViT-G and Classif ViT-e.

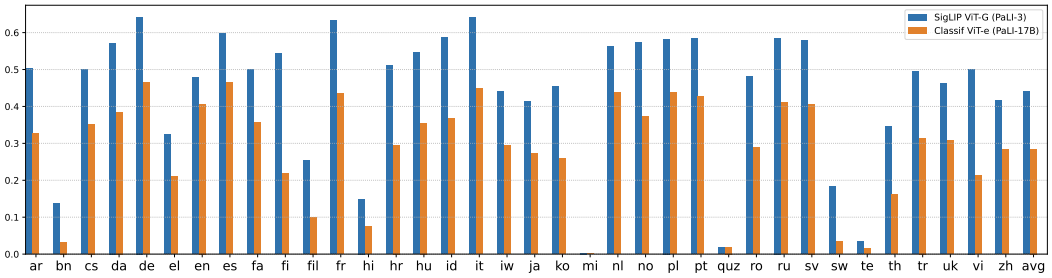


Figure 4: Text-to-image zero-shot retrieval recall@1 on crossmodal-3600 for SigLIP ViT-G and Classif ViT-e.

A 附加结果：视频字幕生成和问答

A.1 数据集与基准

由于在实验时新收集数据时，基准测试中的并非所有视频都可在线获取，因此在某些情况下，有效视频数量少于公开的官方划分数量。表 10 报告了我们在训练和评估中使用的子集的详细数量。我们遵循与 PaLI-X（Chen et al., 2023a）相同的实验设置，包括数据集划分配置和评估指标。更多与数据相关的细节请参阅 PaLI-X（Chen et al., 2023a）。

		MSR-VTT	VATEX	ANet-Cap	SMIT	M-V-QA	. ANet-QA	NExT-QA
Original size	valid.	497	3000	17505	14604	12278	18000	5343
	test	2990	6000	17031	3513	72821	8000	9178
Dataset size	valid.	325	2646	14566	8096	8160	10000	5343
	test	2135	5242	14197	3513	52623	7040	9178
% Remaining	valid.	65.39	88.20	83.21	100.00	66.46	-	100.00
	test	71.40	87.37	83.36	100.00	72.26	88.00	100.00

表10：由于我们新近收集了这些数据集，实际训练数据量小于公开基准，使任务更具挑战性。除 NextQA 和 SMIT 外，训练和评估中均有超过 10% 的视频缺失。

B A 附加结果：CROSSMODAL-3600 检索 评估

本节中，我们展示了在 Crossmodal-3600（Thapliyal 等，2022）上零样本图文检索结果（recall@1）的更多结果。SigLIP ViT-G 和 Classif ViT-e 在 36 种语言上的详细结果见图 3、图 4 和表 11。

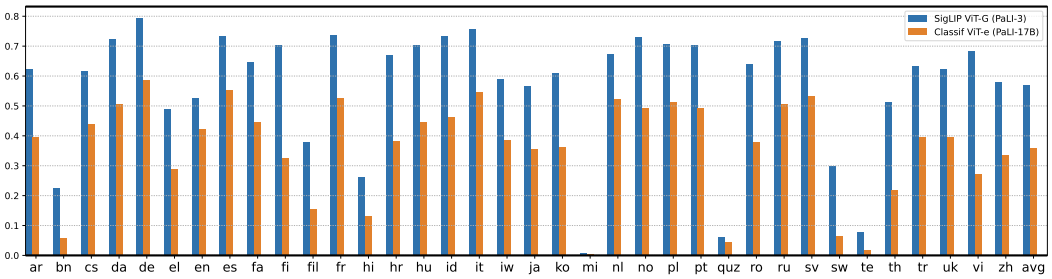


图 3：SigLIP ViT-G 和 Classif ViT-e 在 Crossmodal-3600 上的图像到文本零样本检索 recall@1。

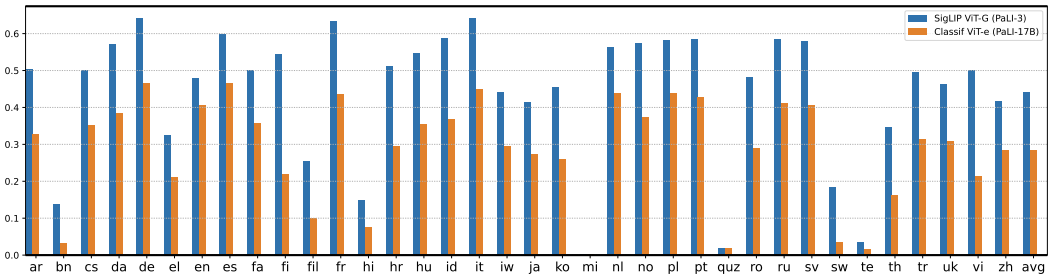


图 4：SigLIP ViT-G 和 Classif ViT-e 在 crossmodal-3600 上的文本到图像零样本检索召回率@1。

Language	Image-to-text		Text-to-image	
	SigLIP ViT-G	Classif ViT-e	SigLIP ViT-G	Classif ViT-e
ar	62.22	39.69	50.33	32.60
bn	22.67	5.67	13.61	3.31
cs	61.69	44.03	50.10	35.24
da	72.47	50.75	57.16	38.48
de	79.28	58.53	64.20	46.50
el	49.11	29.03	32.36	20.92
en	52.64	42.11	47.89	40.63
es	73.31	55.22	59.77	46.55
fa	64.56	44.50	50.09	35.58
fi	70.39	32.64	54.34	21.80
fil	37.89	15.53	25.40	10.04
fr	73.81	52.61	63.28	43.47
hi	26.22	13.14	14.94	7.42
hr	66.89	38.31	51.03	29.55
hu	70.22	44.67	54.63	35.49
id	73.53	46.33	58.62	36.75
it	75.56	54.53	64.14	44.76
iw	59.14	38.67	44.10	29.39
ja	56.69	35.47	41.31	27.24
ko	61.03	36.11	45.50	25.95
mi	0.64	0.33	0.30	0.22
nl	67.25	52.14	56.23	43.79
no	73.03	49.17	57.40	37.35
pl	70.69	51.42	58.24	43.72
pt	70.22	49.19	58.47	42.73
quz	6.28	4.31	1.89	1.90
ro	63.92	37.75	48.20	28.82
ru	71.69	50.64	58.51	41.11
sv	72.69	53.22	57.76	40.66
sw	29.86	6.42	18.31	3.41
te	7.81	1.92	3.39	1.42
th	51.14	22.00	34.65	16.06
tr	63.36	39.50	49.41	31.47
uk	62.19	39.53	46.15	30.81
vi	68.50	27.08	50.14	21.28
zh	57.92	33.61	41.51	28.24
avg	56.85	35.99	43.98	28.46

Table 11: Crossmodal-3600 zero-shot retrieval recall@1 (%) for SigLIP ViT-G (2B params) and Classif ViT-e (4B params). SigLIP ViT-G is significantly better than Classif ViT-e across all the languages. SigLIP improves from 28.5% to 44.0% on average for the text-to-image retrieval task.

Language	Image-to-text		Text-to-image	
	SigLIP ViT-G	Classif ViT-e	SigLIP ViT-G	Classif ViT-e
ar	62.22	39.69	50.33	32.60
bn	22.67	5.67	13.61	3.31
cs	61.69	44.03	50.10	35.24
da	72.47	50.75	57.16	38.48
de	79.28	58.53	64.20	46.50
el	49.11	29.03	32.36	20.92
en	52.64	42.11	47.89	40.63
es	73.31	55.22	59.77	46.55
fa	64.56	44.50	50.09	35.58
fi	70.39	32.64	54.34	21.80
fil	37.89	15.53	25.40	10.04
fr	73.81	52.61	63.28	43.47
hi	26.22	13.14	14.94	7.42
hr	66.89	38.31	51.03	29.55
hu	70.22	44.67	54.63	35.49
id	73.53	46.33	58.62	36.75
it	75.56	54.53	64.14	44.76
iw	59.14	38.67	44.10	29.39
ja	56.69	35.47	41.31	27.24
ko	61.03	36.11	45.50	25.95
mi	0.64	0.33	0.30	0.22
nl	67.25	52.14	56.23	43.79
no	73.03	49.17	57.40	37.35
pl	70.69	51.42	58.24	43.72
pt	70.22	49.19	58.47	42.73
quz	6.28	4.31	1.89	1.90
ro	63.92	37.75	48.20	28.82
ru	71.69	50.64	58.51	41.11
sv	72.69	53.22	57.76	40.66
sw	29.86	6.42	18.31	3.41
te	7.81	1.92	3.39	1.42
th	51.14	22.00	34.65	16.06
tr	63.36	39.50	49.41	31.47
uk	62.19	39.53	46.15	30.81
vi	68.50	27.08	50.14	21.28
zh	57.92	33.61	41.51	28.24
avg	56.85	35.99	43.98	28.46

表 11：SigLIP ViT-G（20 亿参数）和 Classif ViT-e（40 亿参数）在 Crossmodal-3600 零样本检索中的 recall@1（%）。SigLIP ViT-G 在所有语言上均显著优于 Classif ViT-e。在文本到图像检索任务中，SigLIP 的平均表现从 28.5% 提升到 44.0%。